# ENGINEERING THERMODYNAMICS (MME 365)

# PROPERTIES OF LIQUIDS AND VAPOURS

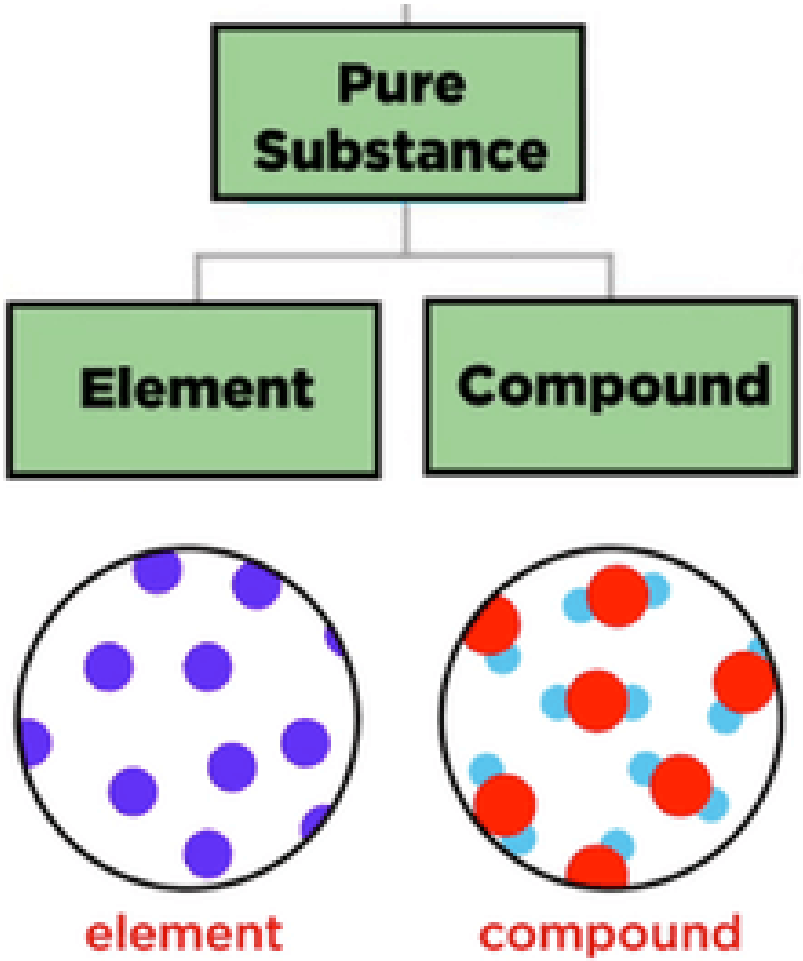
**Kofi Owura Amoabeng (PhD)**

Department of Mechanical Engineering

Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

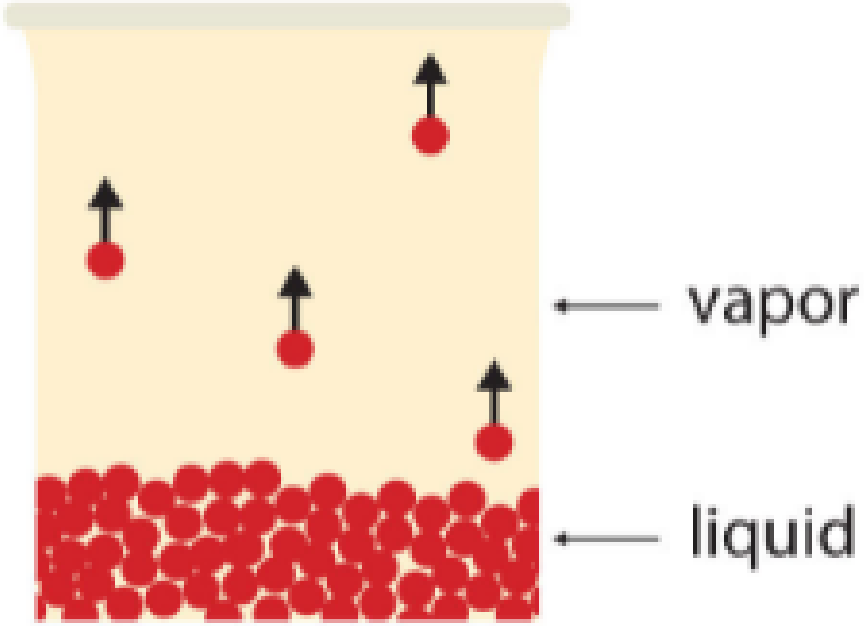
# Pure Substance

- A substance that has a fixed chemical composition throughout is called a **pure substance** such as water, air, and nitrogen.
- A pure substance may exist in different phases. The three principal phases are solid, liquid, and gas.
- A **phase** is defined as having a distinct molecular arrangement that is homogenous throughout and separated from others (if any) by easily identifiable boundary surfaces.
- A pure substance does not have to be of a single element or compound. A mixture of two or more phases of a pure substance is still a pure substance as long as the chemical composition of all phases is the same.



# Properties of Liquids and Vapours

- **Matter may exist in the form of a solid, liquid or a gas.**
- A **vapour** is a gas existing at a temperature at which it can be liquefied by an increase of pressure.
- **State Postulate:** Two independent properties are needed to completely specify the state of a thermodynamic system.
- If these two independent properties are known, the other properties can be determined.

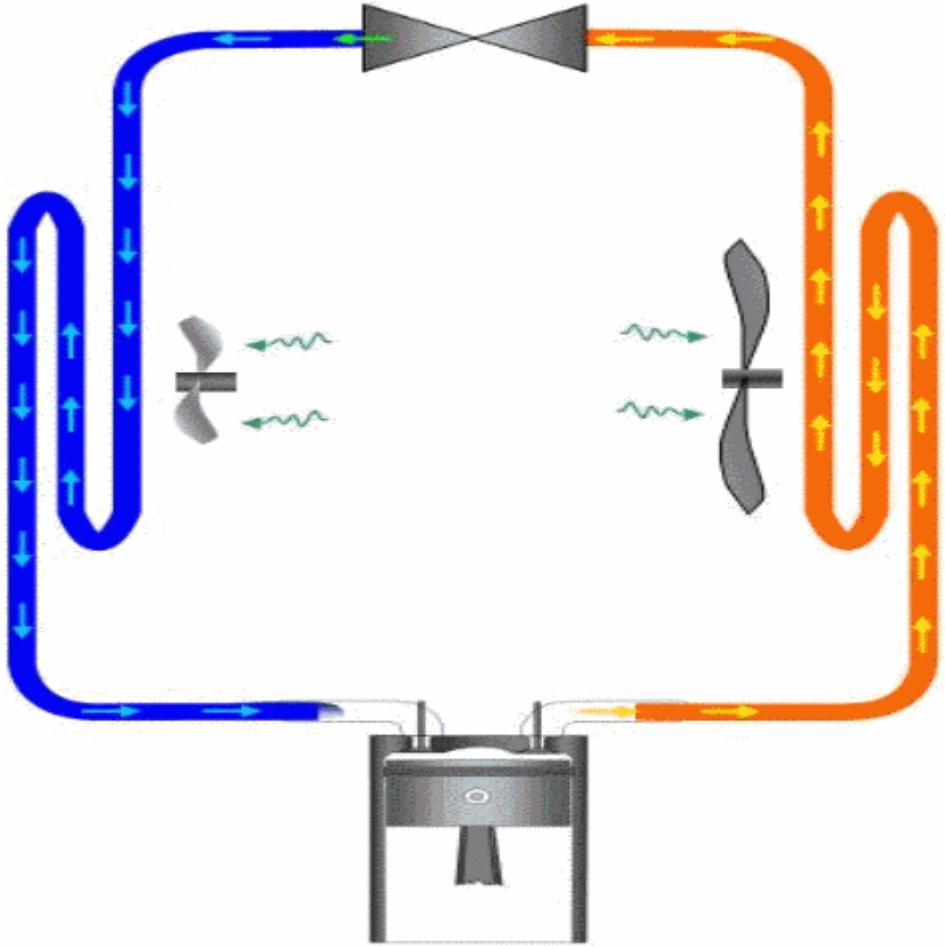




www.knust.edu.gh

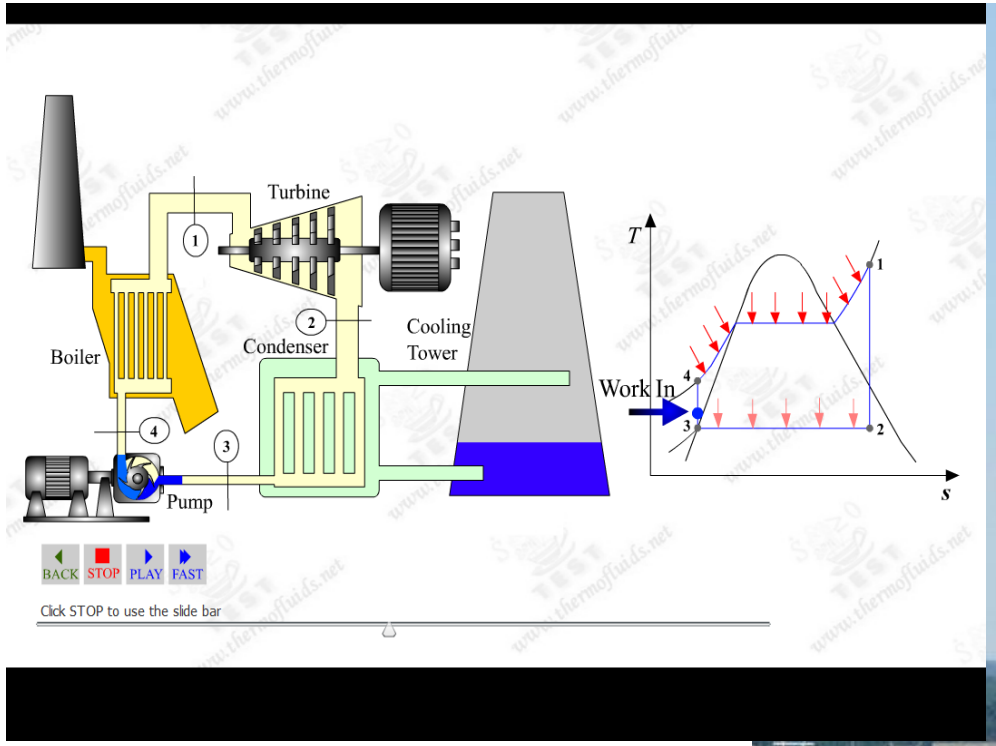
# Properties of Liquids and Vapours

## Refrigeration cycle (Vapour-compression)



# Properties of Liquids and Vapours

## Steam power cycle

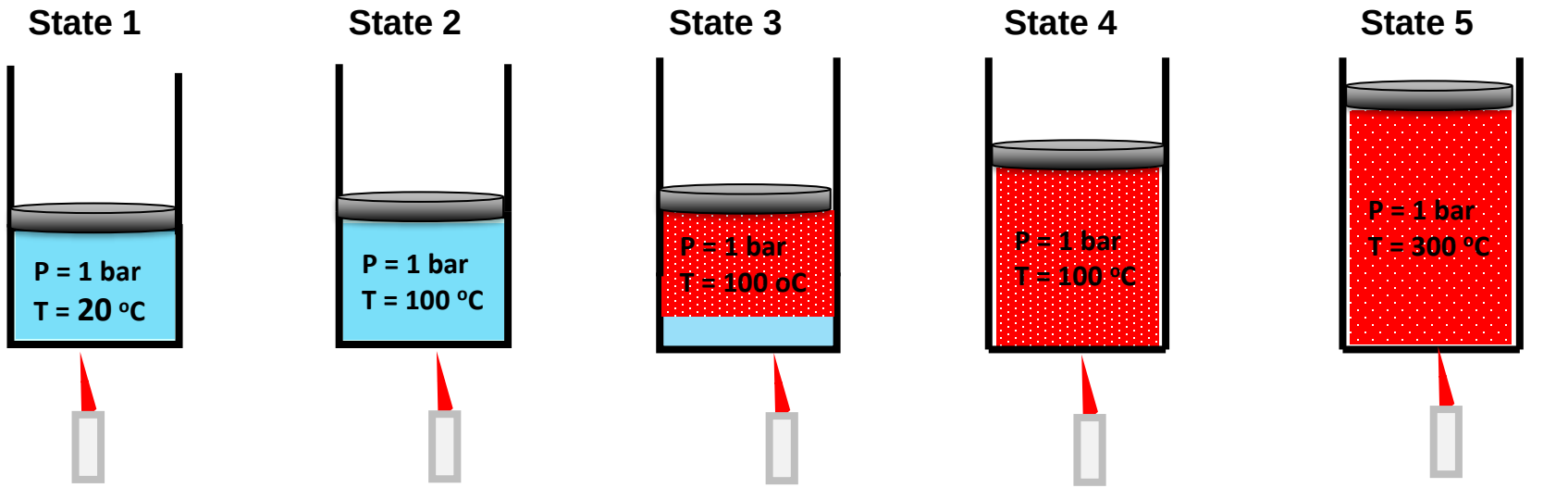





# Properties of Liquids and Vapours

## ❖ Phase Change Experiment

❖ Consider a unit mass (1 kg) of water at a temperature of 20 °C, atmospheric pressure of 1.013 bar, contained in a cylinder as shown.




- **Under-saturated liquid**
- **Sub-cooled Liquid**
- **Compressed liquid**

**Saturated liquid**

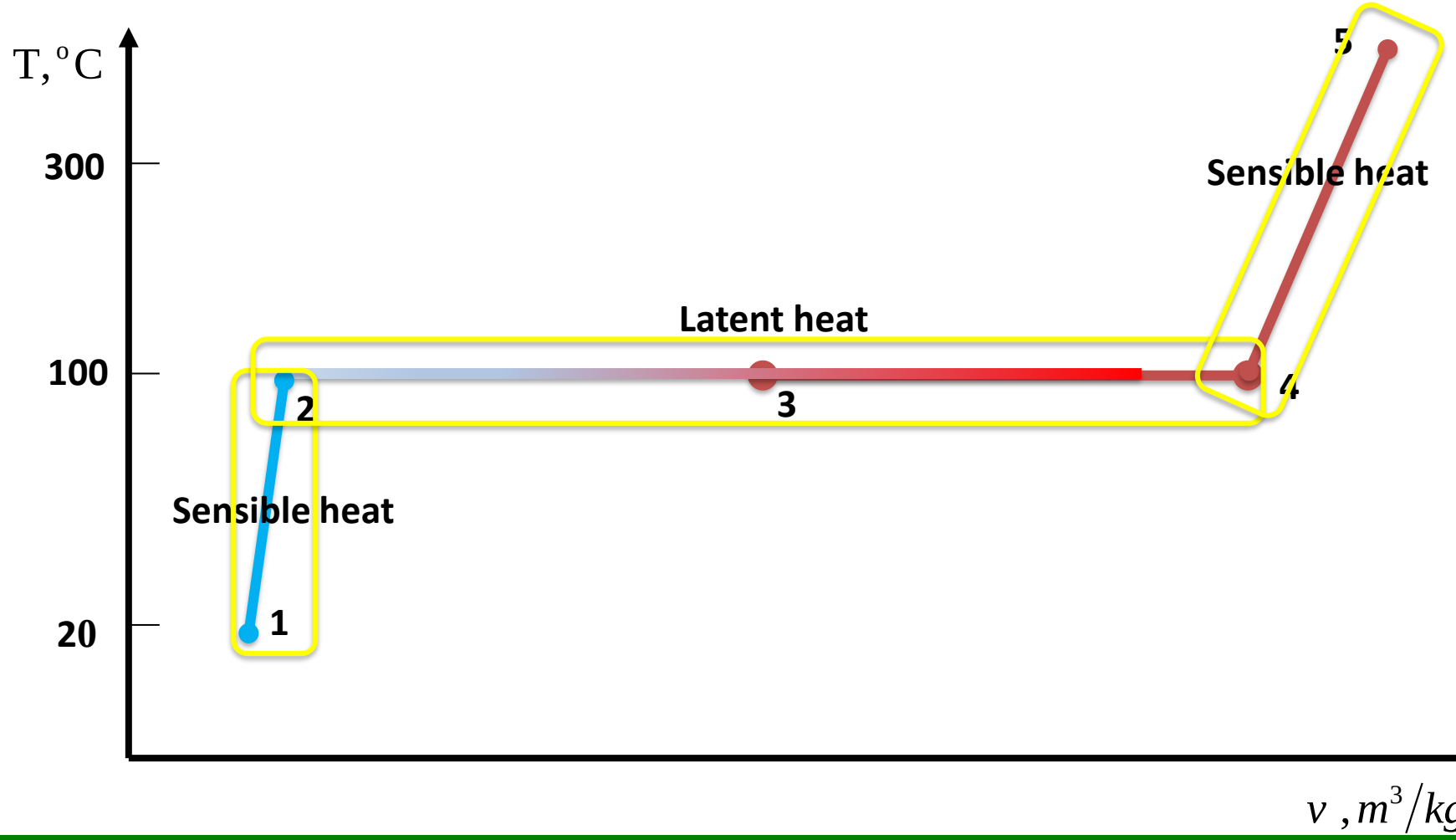
**Saturated liquid-vapour mixture**

**Saturated vapour**

**Superheated vapour**

# Properties of Liquids and Vapours

## ❖ Temperature-Specific Volume Diagram for Water

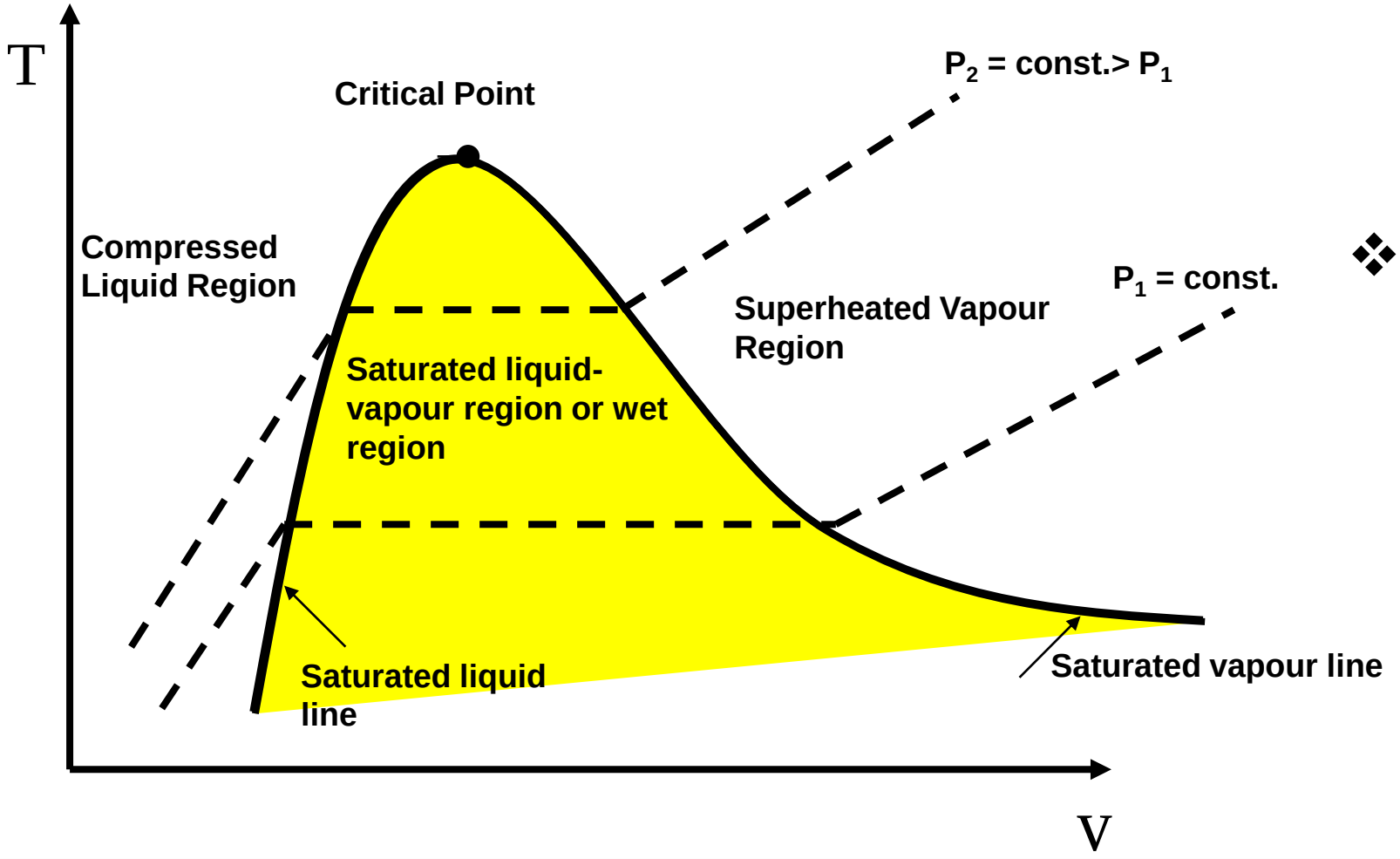





www.knust.edu.gh

# Properties of Liquids and Vapours

## ❖ Temperature-Volume Diagram for Water




- ❖ The properties of saturated liquid, saturated vapour and superheated vapour for water are listed in tables.
- ❖ The **subscript 'f'** is used to denote properties of a **saturated liquid**, and the **subscript 'g'** to denote the properties of **saturated vapour**.


www.knust.edu.gh

# Properties of Liquids and Vapours

## ❖ Properties of a Saturated liquid-vapour mixture (wet mixture)

For a saturated mixture state

$$m = m_f + m_g \qquad \text{and} \qquad V = V_f + V_g$$

$$mv = m_f v_f + m_g v_g \Rightarrow v = \frac{m_f}{m} v_f + \frac{m_g}{m} v_g$$

But we define quality(x) also known as dryness fraction as; $x = \frac{m_g}{m}$

$$v = (1 - x)\, v_f + x v_g = v_f + x\left(v_g - v_f\right) = v_f + x\, v_{fg}$$



www.knust.edu.gh

# Table of Properties for Water

## TABLE A–4

Saturated water—Temperature table (*Concluded*)

| | | Specific volume, $m^3/kg$ | | Internal energy, kJ/kg | | | Enthalpy, kJ/kg | | | Entropy, kJ/kg·K | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Temp., $T$ °C | Sat. press., $P_{sat}$ kPa | Sat. liquid, $v_f$ | Sat. vapor, $v_g$ | Sat. liquid, $u_f$ | Evap., $u_{fg}$ | Sat. vapor, $u_g$ | Sat. liquid, $h_f$ | Evap., $h_{fg}$ | Sat. vapor, $h_g$ | Sat. liquid, $s_f$ | Evap., $s_{fg}$ | Sat. vapor, $s_g$ |
| 305 | 9209.4 | 0.001425 | 0.019932 | 1360.0 | 1195.9 | 2555.8 | 1373.1 | 1366.3 | 2739.4 | 3.3024 | 2.3633 | 5.6657 |
| 310 | 9865.0 | 0.001447 | 0.018333 | 1387.7 | 1159.3 | 2547.1 | 1402.0 | 1325.9 | 2727.9 | 3.3506 | 2.2737 | 5.6243 |
| 315 | 10,556 | 0.001472 | 0.016849 | 1416.1 | 1121.1 | 2537.2 | 1431.6 | 1283.4 | 2715.0 | 3.3994 | 2.1821 | 5.5816 |
| 320 | 11,284 | 0.001499 | 0.015470 | 1445.1 | 1080.9 | 2526.0 | 1462.0 | 1238.5 | 2700.6 | 3.4491 | 2.0881 | 5.5372 |
| 325 | 12,051 | 0.001528 | 0.014183 | 1475.0 | 1038.5 | 2513.4 | 1493.4 | 1191.0 | 2684.3 | 3.4998 | 1.9911 | 5.4908 |
| 330 | 12,858 | 0.001560 | 0.012979 | 1505.7 | 993.5 | 2499.2 | 1525.8 | 1140.3 | 2666.0 | 3.5516 | 1.8906 | 5.4422 |
| 335 | 13,707 | 0.001597 | 0.011848 | 1537.5 | 945.5 | 2483.0 | 1559.4 | 1086.0 | 2645.4 | 3.6050 | 1.7857 | 5.3907 |
| 340 | 14,601 | 0.001638 | 0.010783 | 1570.7 | 893.8 | 2464.5 | 1594.6 | 1027.4 | 2622.0 | 3.6602 | 1.6756 | 5.3358 |
| 345 | 15,541 | 0.001685 | 0.009772 | 1605.5 | 837.7 | 2443.2 | 1631.7 | 963.4 | 2595.1 | 3.7179 | 1.5585 | 5.2765 |
| 350 | 16,529 | 0.001741 | 0.008806 | 1642.4 | 775.9 | 2418.3 | 1671.2 | 892.7 | 2563.9 | 3.7788 | 1.4326 | 5.2114 |
| 355 | 17,570 | 0.001808 | 0.007872 | 1682.2 | 706.4 | 2388.6 | 1714.0 | 812.9 | 2526.9 | 3.8442 | 1.2942 | 5.1384 |
| 360 | 18,666 | 0.001895 | 0.006950 | 1726.2 | 625.7 | 2351.9 | 1761.5 | 720.1 | 2481.6 | 3.9165 | 1.1373 | 5.0537 |
| 365 | 19,822 | 0.002015 | 0.006009 | 1777.2 | 526.4 | 2303.6 | 1817.2 | 605.5 | 2422.7 | 4.0004 | 0.9489 | 4.9493 |
| 373.95 | 22,064 | 0.003106 | 0.003106 | 2015.7 | 0 | 2015.7 | 2084.3 | 0 | 2084.3 | 4.4070 | 0 | 4.4070 |
| 90 | 70.183 | 0.001036 | 2.3593 | 376.97 | 2117.0 | 2494.0 | 377.04 | 2282.5 | 2659.6 | 1.1929 | 6.2853 | 7.4782 |
| 95 | 84.609 | 0.001040 | 1.9808 | 398.00 | 2102.0 | 2500.1 | 398.09 | 2269.6 | 2667.6 | 1.2504 | 6.1647 | 7.4151 |
| 100 | 101.42 | 0.001043 | 1.6720 | 419.06 | 2087.0 | 2506.0 | 419.17 | 2256.4 | 2675.6 | 1.3072 | 6.0470 | 7.3542 |
| 105 | 120.90 | 0.001047 | 1.4186 | 440.15 | 2071.8 | 2511.9 | 440.28 | 2243.1 | 2683.4 | 1.3634 | 5.9319 | 7.2952 |
| 110 | 143.38 | 0.001052 | 1.2094 | 461.27 | 2056.4 | 2517.7 | 461.42 | 2229.7 | 2691.1 | 1.4188 | 5.8193 | 7.2382 |
| 115 | 169.18 | 0.001056 | 1.0360 | 482.42 | 2040.9 | 2523.3 | 482.59 | 2216.0 | 2698.6 | 1.4737 | 5.7092 | 7.1829 |
| 120 | 198.67 | 0.001060 | 0.89133 | 503.60 | 2025.3 | 2528.9 | 503.81 | 2202.1 | 2706.0 | 1.5279 | 5.6013 | 7.1292 |

## TABLE A–5

Saturated water—Pressure table (*Concluded*)

| | | Specific volume, $m^3/kg$ | | Internal energy, kJ/kg | | | Enthalpy, kJ/kg | | | Entropy, kJ/kg·K | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Press., $P$ kPa | Sat. temp., $T_{sat}$ °C | Sat. liquid, $v_f$ | Sat. vapor, $v_g$ | Sat. liquid, $u_f$ | Evap., $u_{fg}$ | Sat. vapor, $u_g$ | Sat. liquid, $h_f$ | Evap., $h_{fg}$ | Sat. vapor, $h_g$ | Sat. liquid, $s_f$ | Evap., $s_{fg}$ | Sat. vapor, $s_g$ |
| 13,000 | 330.85 | 0.001566 | 0.012781 | 1511.0 | 985.5 | 2496.6 | 1531.4 | 1131.3 | 2662.7 | 3.5606 | 1.8730 | 5.4336 |
| 14,000 | 336.67 | 0.001610 | 0.011487 | 1548.4 | 928.7 | 2477.1 | 1571.0 | 1067.0 | 2637.9 | 3.6232 | 1.7497 | 5.3728 |
| 15,000 | 342.16 | 0.001657 | 0.010341 | 1585.5 | 870.3 | 2455.7 | 1610.3 | 1000.5 | 2610.8 | 3.6848 | 1.6261 | 5.3108 |
| 16,000 | 347.36 | 0.001710 | 0.009312 | 1622.6 | 809.4 | 2432.0 | 1649.9 | 931.1 | 2581.0 | 3.7461 | 1.5005 | 5.2466 |
| 17,000 | 352.29 | 0.001770 | 0.008374 | 1660.2 | 745.1 | 2405.4 | 1690.3 | 857.4 | 2547.7 | 3.8082 | 1.3709 | 5.1791 |
| 18,000 | 356.99 | 0.001840 | 0.007504 | 1699.1 | 675.9 | 2375.0 | 1732.2 | 777.8 | 2510.0 | 3.8720 | 1.2343 | 5.1064 |
| 19,000 | 361.47 | 0.001926 | 0.006677 | 1740.3 | 598.9 | 2339.2 | 1776.8 | 689.2 | 2466.0 | 3.9396 | 1.0860 | 5.0256 |
| 20,000 | 365.75 | 0.002038 | 0.005862 | 1785.8 | 509.0 | 2294.8 | 1826.6 | 585.5 | 2412.1 | 4.0146 | 0.9164 | 4.9310 |
| 21,000 | 369.83 | 0.002207 | 0.004994 | 1841.6 | 391.9 | 2233.5 | 1888.0 | 450.4 | 2338.4 | 4.1071 | 0.7005 | 4.8076 |
| 22,000 | 373.71 | 0.002703 | 0.003644 | 1951.7 | 140.8 | 2092.4 | 2011.1 | 161.5 | 2172.6 | 4.2942 | 0.2496 | 4.5439 |
| 22,064 | 373.95 | 0.003106 | 0.003106 | 2015.7 | 0 | 2015.7 | 2084.3 | 0 | 2084.3 | 4.4070 | 0 | 4.4070 |
| 20 | 60.06 | 0.001017 | 7.6481 | 251.40 | 2204.6 | 2456.0 | 251.42 | 2357.5 | 2608.9 | 0.8320 | 7.0752 | 7.9073 |
| 25 | 64.96 | 0.001020 | 6.2034 | 271.93 | 2190.4 | 2462.4 | 271.96 | 2345.5 | 2617.5 | 0.8932 | 6.9370 | 7.8302 |
| 30 | 69.09 | 0.001022 | 5.2287 | 289.24 | 2178.5 | 2467.7 | 289.27 | 2335.3 | 2624.6 | 0.9441 | 6.8234 | 7.7675 |
| 40 | 75.86 | 0.001026 | 3.9933 | 317.58 | 2158.8 | 2476.3 | 317.62 | 2318.4 | 2636.1 | 1.0261 | 6.6430 | 7.6691 |
| 50 | 81.32 | 0.001030 | 3.2403 | 340.49 | 2142.7 | 2483.2 | 340.54 | 2304.7 | 2645.2 | 1.0912 | 6.5019 | 7.5931 |


E. W. KANDE

## TABLE A–6

### Superheated water

| $T$ °C | $v$ m³/kg | $u$ kJ/kg | $h$ kJ/kg | $s$ kJ/kg·K | $v$ m³/kg | $u$ kJ/kg | $h$ kJ/kg | $s$ kJ/kg·K | $v$ m³/kg | $u$ kJ/kg | $h$ kJ/kg | $s$ kJ/kg·K |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $P$ = 0.01 MPa (45.81°C)* | | | | $P$ = 0.05 MPa (81.32°C) | | | | $P$ = 0.10 MPa (99.61°C) | | | |
| Sat.† | 14.670 | 2437.2 | 2583.9 | 8.1488 | 3.2403 | 2483.2 | 2645.2 | 7.5931 | 1.6941 | 2505.6 | 2675.0 | 7.3589 |
| 50 | 14.867 | 2443.3 | 2592.0 | 8.1741 | | | | | | | | |
| 100 | 17.196 | 2515.5 | 2687.5 | 8.4489 | 3.4187 | 2511.5 | 2682.4 | 7.6953 | 1.6959 | 2506.2 | 2675.8 | 7.3611 |
| 150 | 19.513 | 2587.9 | 2783.0 | 8.6893 | 3.8897 | 2585.7 | 2780.2 | 7.9413 | 1.9367 | 2582.9 | 2776.6 | 7.6148 |
| 200 | 21.826 | 2661.4 | 2879.6 | 8.9049 | 4.3562 | 2660.0 | 2877.8 | 8.1592 | 2.1724 | 2658.2 | 2875.5 | 7.8356 |
| 250 | 24.136 | 2736.1 | 2977.5 | 9.1015 | 4.8206 | 2735.1 | 2976.2 | 8.3568 | 2.4062 | 2733.9 | 2974.5 | 8.0346 |
| 300 | 26.446 | 2812.3 | 3076.7 | 9.2827 | 5.2841 | 2811.6 | 3075.8 | 8.5387 | 2.6389 | 2810.7 | 3074.5 | 8.2172 |
| 400 | 31.063 | 2969.3 | 3280.0 | 9.6094 | 6.2094 | 2968.9 | 3279.3 | 8.8659 | 3.1027 | 2968.3 | 3278.6 | 8.5452 |
| 500 | 35.680 | 3132.9 | 3489.7 | 9.8998 | 7.1338 | 3132.6 | 3489.3 | 9.1566 | 3.5655 | 3132.2 | 3488.7 | 8.8362 |
| 600 | 40.296 | 3303.3 | 3706.3 | 10.1631 | 8.0577 | 3303.1 | 3706.0 | 9.4201 | 4.0279 | 3302.8 | 3705.6 | 9.0999 |
| 700 | 44.911 | 3480.8 | 3929.9 | 10.4056 | 8.9813 | 3480.6 | 3929.7 | 9.6626 | 4.4900 | 3480.4 | 3929.4 | 9.3424 |
| 800 | 49.527 | 3665.4 | 4160.6 | 10.6312 | 9.9047 | 3665.2 | 4160.4 | 9.8883 | 4.9519 | 3665.0 | 4160.2 | 9.5682 |
| 900 | 54.143 | 3856.9 | 4398.3 | 10.8429 | 10.8280 | 3856.8 | 4398.2 | 10.1000 | 5.4137 | 3856.7 | 4398.0 | 9.7800 |
| 1000 | 58.758 | 4055.3 | 4642.8 | 11.0429 | 11.7513 | 4055.2 | 4642.7 | 10.3000 | 5.8755 | 4055.0 | 4642.6 | 9.9800 |
| 1100 | 63.373 | 4260.0 | 4893.8 | 11.2326 | 12.6745 | 4259.9 | 4893.7 | 10.4897 | 6.3372 | 4259.8 | 4893.6 | 10.1698 |
| 1200 | 67.989 | 4470.9 | 5150.8 | 11.4132 | 13.5977 | 4470.8 | 5150.7 | 10.6704 | 6.7988 | 4470.7 | 5150.6 | 10.3504 |
| 1300 | 72.604 | 4687.4 | 5413.4 | 11.5857 | 14.5209 | 4687.3 | 5413.3 | 10.8429 | 7.2605 | 4687.2 | 5413.3 | 10.5229 |


www.knust.edu.gh

# Use of Interpolation in Tables of Properties

Determine the temp. of water at a state of P = 0.5 MPa and h = 2970 kJ/kg

**TABLE A–6**

Superheated water

| T °C | v m³/kg | u kJ/kg | h kJ/kg | s kJ/kg·K | v m³/kg | u kJ/kg | h kJ/kg | s kJ/kg·K | v m³/kg | u kJ/kg | h kJ/kg | s kJ/kg·K |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | P = 0.50 MPa (151.83°C) | | | | P = 0.60 MPa (158.83°C) | | | | P = 0.80 MPa (170.41°C) | | | |
| Sat. | 0.37483 | 2560.7 | 2748.1 | 6.8207 | 0.31560 | 2566.8 | 2756.2 | 6.7593 | 0.24035 | 2576.0 | 2768.3 | 6.6616 |
| 200 | 0.42503 | 2643.3 | 2855.8 | 7.0610 | 0.35212 | 2639.4 | 2850.6 | 6.9683 | 0.26088 | 2631.1 | 2839.8 | 6.8177 |
| 250 | 0.47443 | 2723.8 | 2961.0 | 7.2725 | 0.39390 | 2721.2 | 2957.6 | 7.1833 | 0.29321 | 2715.9 | 2950.4 | 7.0402 |
| 300 | 0.52261 | 2803.3 | 3064.6 | 7.4614 | 0.43442 | 2801.4 | 3062.0 | 7.3740 | 0.32416 | 2797.5 | 3056.9 | 7.2345 |
| 350 | 0.57015 | 2883.0 | 3168.1 | 7.6346 | 0.47428 | 2881.6 | 3166.1 | 7.5481 | 0.35442 | 2878.6 | 3162.2 | 7.4107 |
| 400 | 0.61731 | 2963.7 | 3272.4 | 7.7956 | 0.51374 | 2962.5 | 3270.8 | 7.7097 | 0.38429 | 2960.2 | 3267.7 | 7.5735 |
| 500 | 0.71095 | 3129.0 | 3484.5 | 8.0893 | 0.59200 | 3128.2 | 3483.4 | 8.0041 | 0.44332 | 3126.6 | 3481.3 | 7.8692 |
| 600 | 0.80409 | 3300.4 | 3702.5 | 8.3544 | 0.66976 | 3299.8 | 3701.7 | 8.2695 | 0.50186 | 3298.7 | 3700.1 | 8.1354 |
| 700 | 0.89696 | 3478.6 | 3927.0 | 8.5978 | 0.74725 | 3478.1 | 3926.4 | 8.5132 | 0.56011 | 3477.2 | 3925.3 | 8.3794 |

| T (°C) | h (kJ/kg) |
|---|---|
| 250 | 2961.0 |
| 254.33 | 2970.0 |
| 300 | 3064.6 |

$$\frac{\text{———}}{\text{———}} = \frac{\text{———}}{\text{———}}$$

$$\frac{300 - ?}{? - 250} = \frac{3064.6 - 2970.0}{2970.0 - 2961.0}$$

# Properties of Liquids and Vapours: Problems

## Problem 1

Steam at 3 bar has a dryness fraction of 0.8. Calculate its specific volume, specific enthalpy, and specific internal energy

**Suggested Answers:**

**[0.4848 $m^3/kg$; 2292.2 kJ/kg; 2147.4 kJ/kg]**

## Problem 2

A cylinder contains 0.34 kg of steam at 10 bar. If the volume of the cylinder is 0.05 $m^3$, calculate the dryness fraction of the steam, the internal energy, and the enthalpy.

**Suggested Answers:**

**[0.755; 726 kJ, 776 kJ]**

# Properties of Liquids and Vapours: Problems

## Problem 3

A mass of 200 g of saturated liquid water is completely vaporized at a constant pressure of 100 kPa. Determine the amount of heat energy added to the water.

**Suggested Answer: [451.5 kJ]**

## Problem 4

10 kg of a mixture of saturated vapour and saturated liquid in a 5 $m^3$ tank is at a pressure of 150 kPa. Find its temperature and the fraction of vapour in the mixture.

**Suggested Answers:**

**[111.35 °C; 0.43]**

# Properties of Liquids and Vapours: Problems

## Problem 5

0.05 kg of Steam at 100 kPa, dryness fraction of 0.85 is heated at constant pressure to a temperature of 200 °C. Determine the changes in volume, internal energy, enthalpy and entropy of the steam.

**Suggested Answer: [0.03662 $m^3$; 23.3 kJ; 26.96 kJ; 0.0693 kJ/K]**

## Problem 6

1 kg of steam 0.8 dry at 9 bar expands during a non-flow polytropic process according to the law $PV^{1.1}$ = constant until the pressure becomes 3 bar. Calculate the final dryness fraction, the work energy transfer, and the heat energy transfer.

**Suggested Answer: [0.772; 144.9 kJ; 56.7 kJ]**


www.knust.edu.gh

# Properties of Liquids and Vapours: Problems

## Problem 7

Steam enters a steam turbine at 5 MPa, and 500 °C at a rate of 50 kg/s and leaves it at a pressure of 10 kPa. The quality of steam at the turbine exit is 0.95. What is the magnitude of the power output from this turbine?

**Suggested Answer: [48.5 MW]**

## Problem 8

Steam is accelerated by a nozzle steadily from a low velocity to 210 m/s at a rate of 3.2 kg/s. If the temperature and pressure of the steam at the nozzle exit are 400 °C and 2 MPa, what is the exit area of the nozzle in $cm^2$?

**Suggested Answer: [23.04 $cm^2$]**

# Properties of Liquids and Vapours: Problems

## Problem 9

Steam at 1MPa and 300 °C is throttled adiabatically to a pressure of 0.4 MPa. If the change in kinetic energy is negligible, what is the specific volume of the steam after throttling?

**Suggested Answer: [0.646 $m^3/kg$;]**

## Problem 10

Steam expands in a turbine from 4 MPa and 500 °C to 0.5 MPa and 250 °C at a rate of 1350 kg/h. Heat is lost from the turbine at a rate of 25 kJ/s during the process. What is the power output of the turbine?

**Suggested Answer: [156.9 kW]**


www.knust.edu.gh

# Properties of Liquids and Vapours: Problems

## Problem 11

The power output of an adiabatic steam turbine is **6 MW**, and the inlet and the exit conditions of the steam are given as

| Inlet | Outlet |
|---|---|
| $P_1 = 3.5$ MPa | $P_2 = 20$ kPa |
| $T_1 = 600^oC$ | $x_2 = 92\%$ |
| $C_1 = 50$ m/s | $C_2 = 150$ m/s |
| $Z_1 = 10$ m | $Z_2 = 5$ m |

Calculate the mass flow rate of the steam in kg/s

**Suggested Answer: [4.81 kg/s]**



www.knust.edu.gh


THANK
YOU